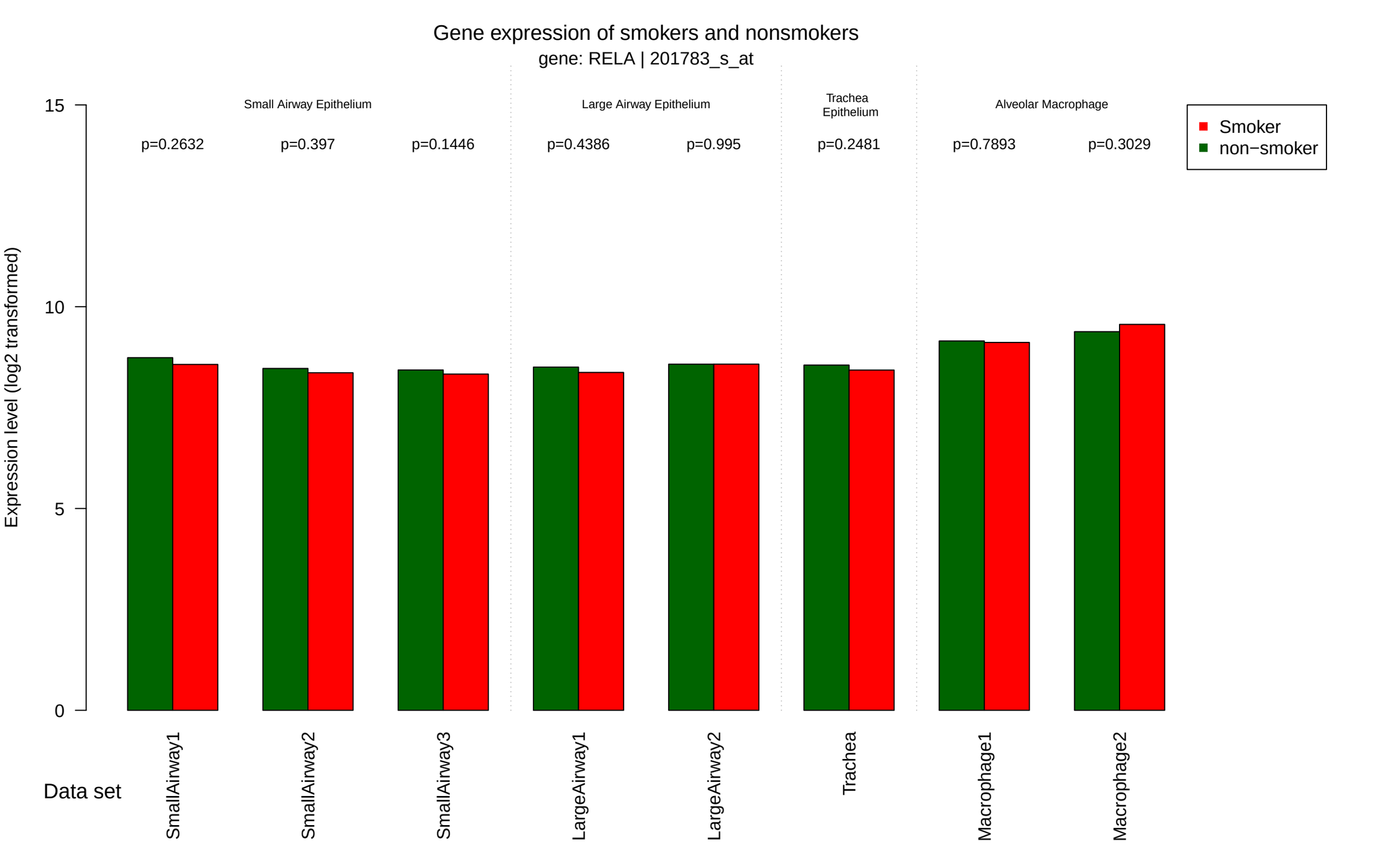
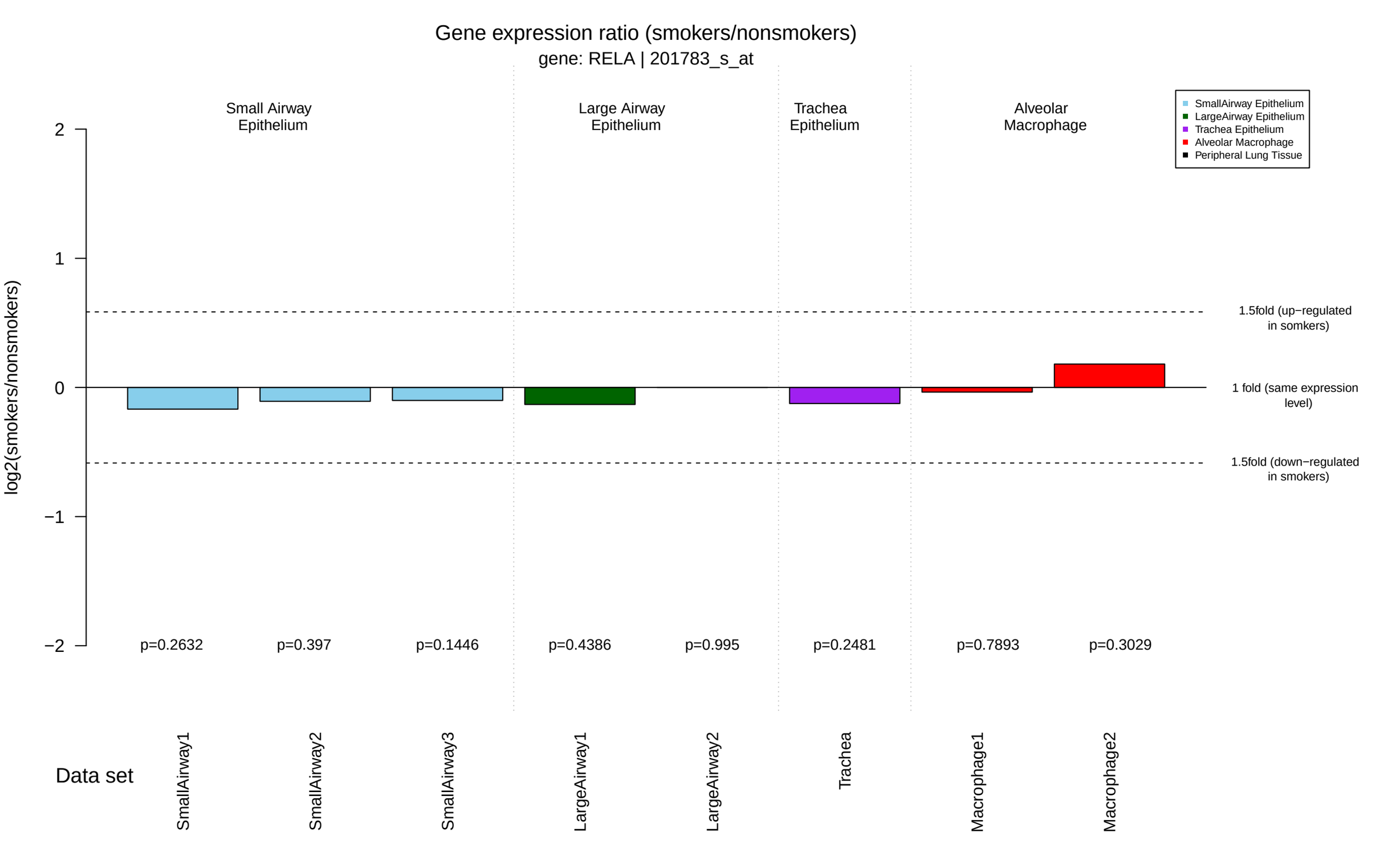
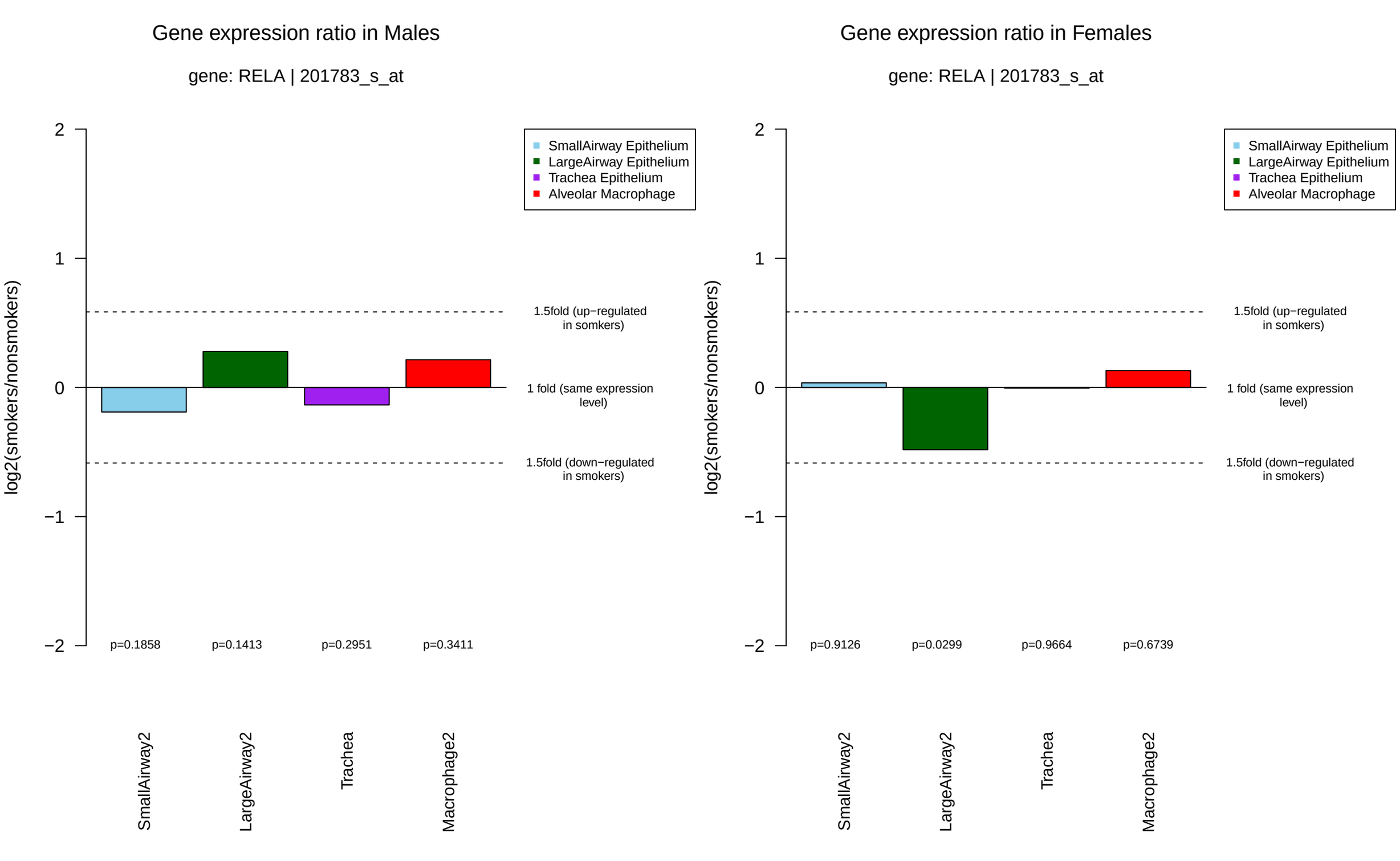
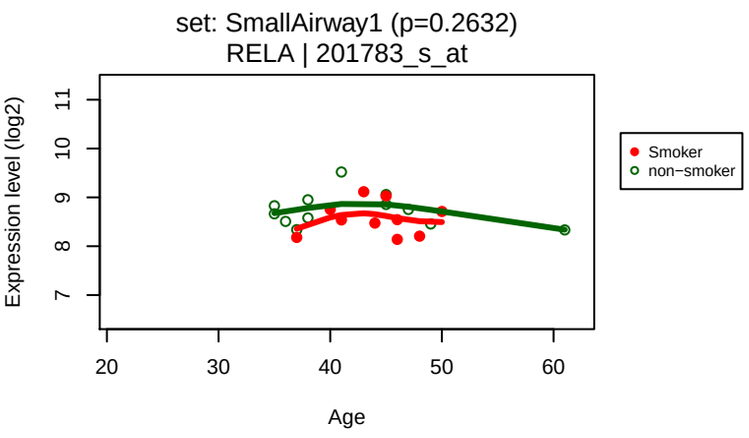
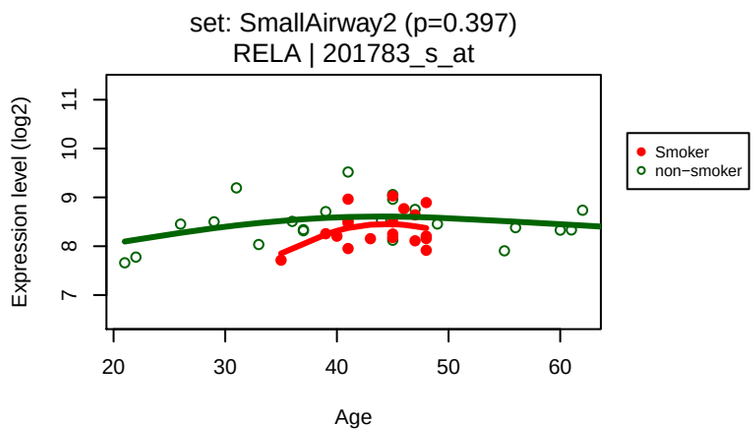
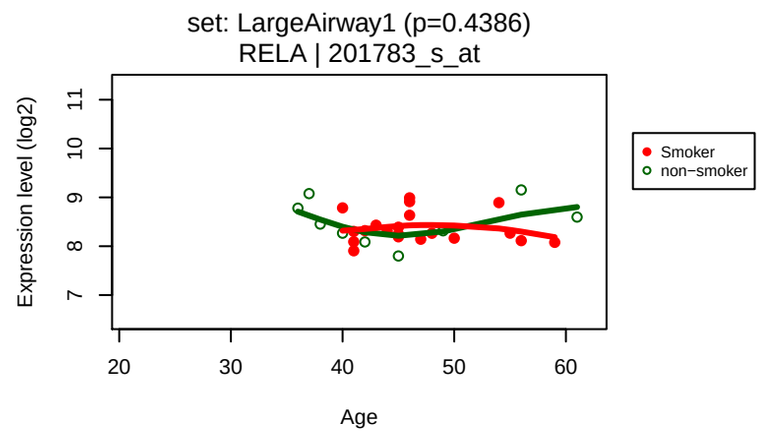

Gene expression of smokers and nonsmokers
gene: RELA | 201783_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2632
p=0.397
p=0.1446
p=0.4386
p=0.995
p=0.2481
p=0.7893
p=0.3029
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

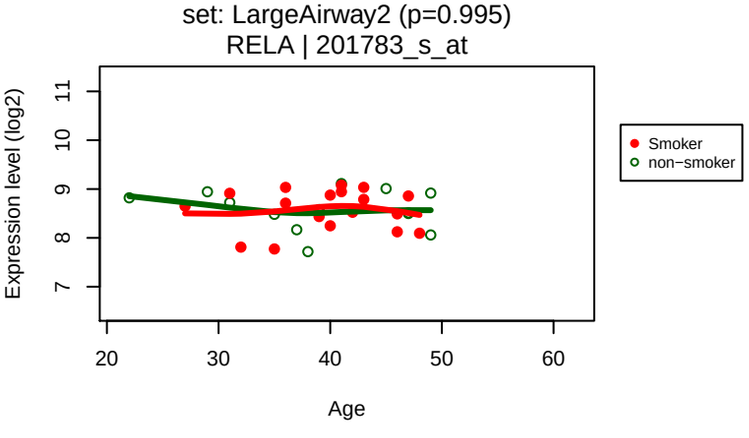

Gene expression ratio (smokers/nonsmokers)
gene: RELA | 201783_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2632
p=0.397
p=0.1446
p=0.4386
p=0.995
p=0.2481
p=0.7893
p=0.3029
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

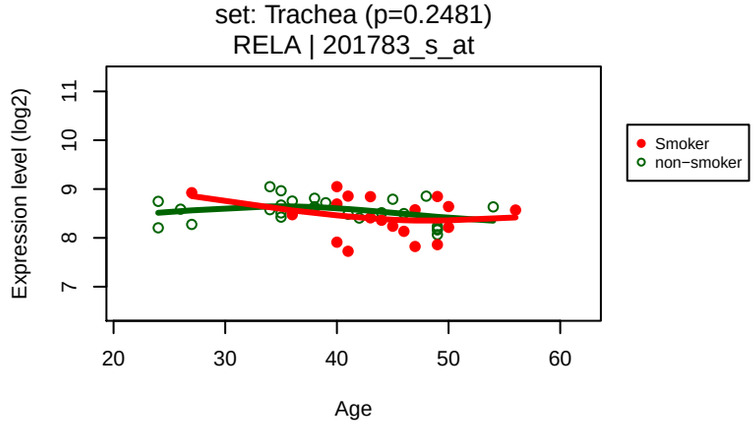

Gene expression ratio in Males
gene: RELA | 201783_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1858
p=0.1413
p=0.2951
p=0.3411
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RELA | 201783_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9126
p=0.0299
p=0.9664
p=0.6739
SmallAirway2
LargeAirway2
Trachea
Macrophage2

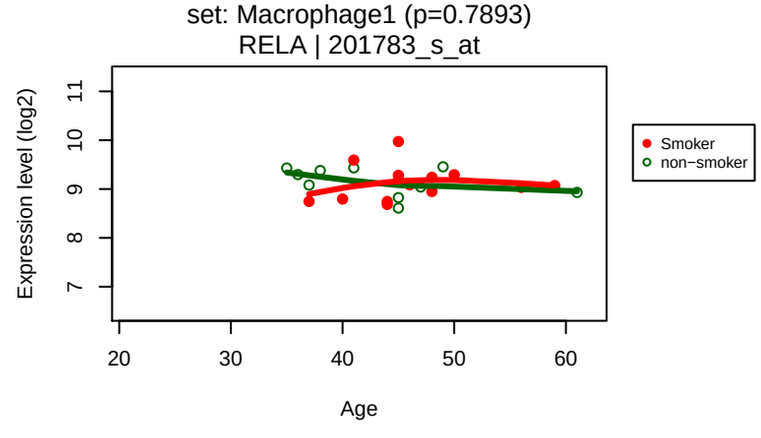

set: SmallAirway1 (p=0.2632)
RELA | 201783_s_at
Expression level (log2)
Age
Smoker
non−smoker

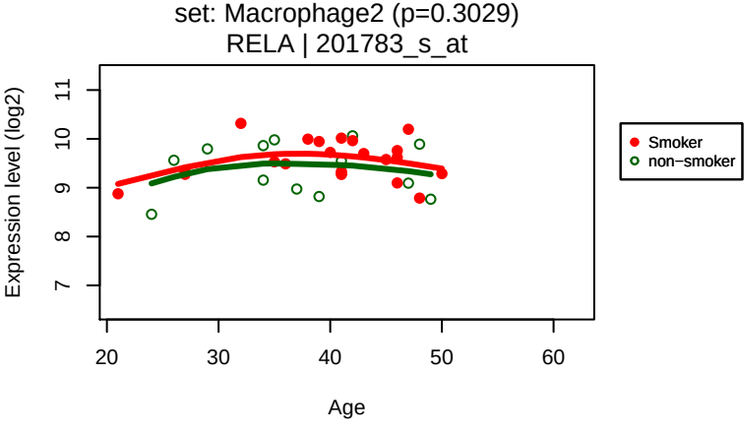

set: SmallAirway2 (p=0.397)
RELA | 201783_s_at
Expression level (log2)
Age
Smoker
non−smoker

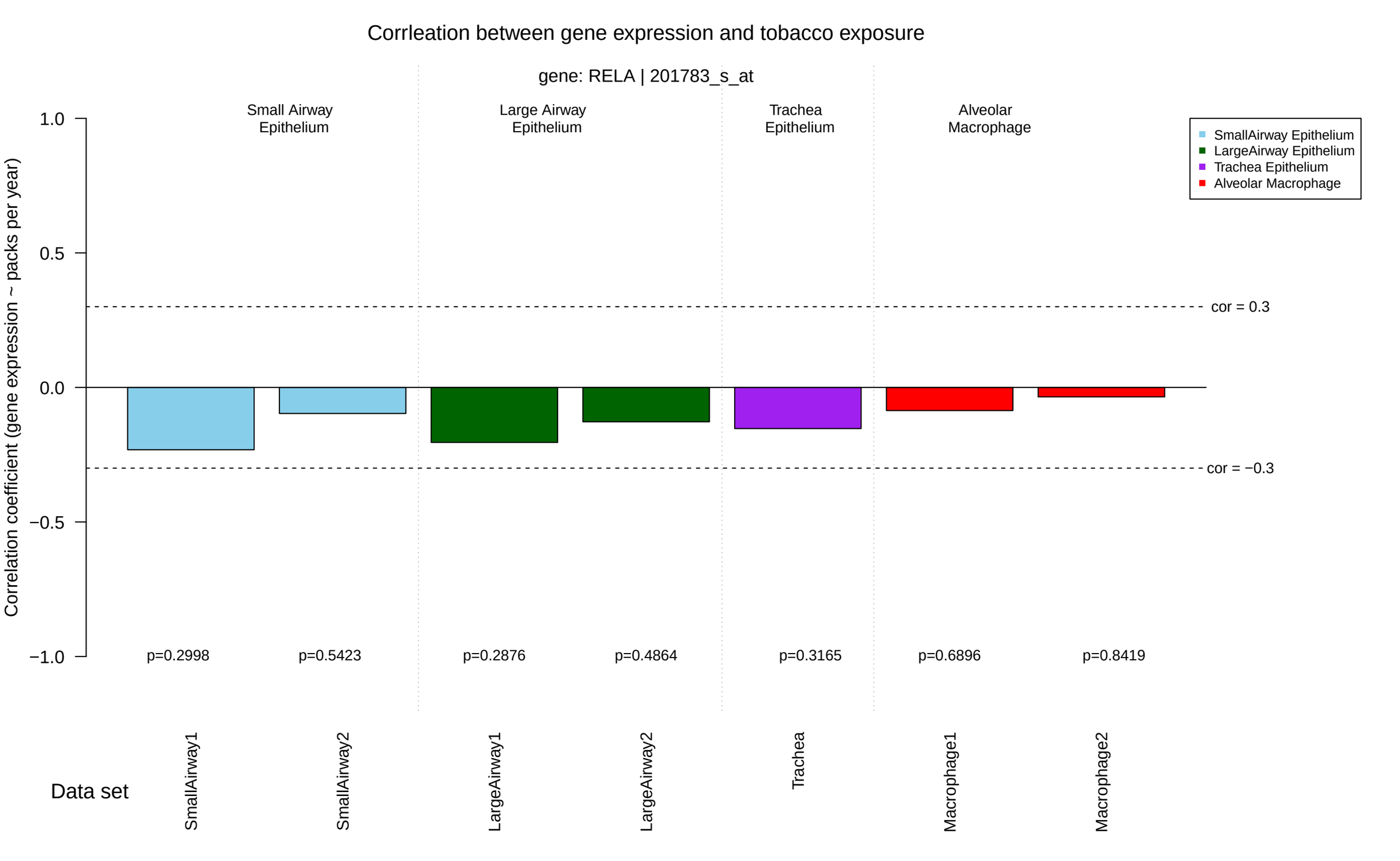

set: LargeAirway1 (p=0.4386)
RELA | 201783_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.995)
RELA | 201783_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2481)
RELA | 201783_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7893)
RELA | 201783_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3029)
RELA | 201783_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RELA | 201783_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2998
p=0.5423
p=0.2876
p=0.4864
p=0.3165
p=0.6896
p=0.8419
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2